

# A sharp moment-theoretic bridge between sampling and node-and-edge convergence

Candidate partial result for arXiv:1406.7846

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**Status.** Candidate substantial partial result, likely valid. For genuine probability-valued multigraphons, sampling convergence implies full node-and-edge density convergence exactly when the one-edge bond moments converge. Conversely, density convergence implies sampling convergence when the limiting one-edge moment sequence is Stieltjes-determinate. In particular, the notions are equivalent under one uniform exponential-moment bound. Counterexamples show that uniform integrability and moment determinacy are the two genuine obstructions.

## 1 Source question

Kunszenti-Kovács, Lovász, and Szegedy study multigraph limits with unbounded edge multiplicities in [1]. In Section 4.4 they recall that homomorphism-density convergence and finite-sample convergence coincide for bounded or compact decorations and then ask what happens for unbounded multigraphs. They isolate the unboundedness of the density tests and moment indeterminacy as obstacles, give a counterexample in full generality, and conclude that they expect settings where the notions coincide but do not investigate them.

The question and its two stated obstructions appear on source PDF page 32:

our formalism, following Example 2.4, this means that given a sequence  $W_n : [0, 1]^2 \rightarrow \mathbb{R}^2$  of symmetric measurable functions with  $W_n(x, y) \in \{(a, b) \in \mathbb{R}^2 \mid a, b \geq 0, a + b = 1\}$ , we have that  $(t(F, W_n))_{n \in \mathbb{N}^+}$  converges for every simple graph  $F$  if and only if the distribution of  $\mathbb{G}^{\mathfrak{R}}(k, W_n)$  over the set of simple graphs on  $k$  vertices converges in the weak-\* (or vague) topology – here equivalent to weak convergence of measures – as  $n \rightarrow \infty$  for every  $k \in \mathbb{N}^*$ . In more generality, Lovász and Szegedy showed ([11, Theorem 2.3]) the same equivalence in the setting of compact decorations, i.e., when  $\mathcal{B} = C(\mathcal{K})$  for some compact separable topological space  $\mathcal{K}$  (see Section 2.3). It is thus a natural question to ask what happens in the case of multigraphs with unbounded edge multiplicities.

Convergence of the samples would provide a connection to exchangeable arrays and Aldous’ representation theorem for these.

The first non-trivial hurdle is what topology to choose for the distributions. As  $\mathbb{G}^{\mathfrak{R}}(k, W_n)$  now lives on a locally compact, rather than a compact space, weak convergence (from the weak duality with  $C_b(X)$ ) and weak-\* convergence (from the Riesz-Markov duality ( $C_0(X)^* = M(X)$ )). The second is that the former convergence notion would preserve total measure (i.e. a sequence of probability distributions would have a limit that’s always a probability distribution), but compactness is lost, whereas the latter allows one to make use of weak-\* compactness, but a sequence of probability distributions may actually end up with a zero limit.

Thirdly, no matter which we choose, even if we assume that the sequences  $\mathbb{G}^{\mathfrak{R}}(k, W_n)$  have probability distributions as their limits, the density functions  $t(\mathbf{F}, \cdot)$  are not bounded (except for zero-decorations), and even less in  $C_0$ , convergence of distributions does not immediately imply convergence in densities.

Issues are also present for the reverse direction, in that unboundedness naturally leads into moment indeterminacy problems. In particular the counter-example from [8, Section 7.3] can be adapted to this setting as well.

The explicit expected-settings sentence is on source PDF page 33:

We do expect to have settings (as in the previous section) where these convergence notions coincide, but it is beyond the scope of the present paper to go investigate the details.

The theorem below identifies the two obstacles exactly enough to give a broad positive regime and two sharp failure examples.

## 2 Framework and the mixed-moment identity

A *probability multigraphon* is a symmetric measurable kernel

$$W : [0, 1]^2 \longrightarrow \mathcal{P}(\mathbb{N}_0).$$

For the random  $k$ -vertex sample  $\mathbb{G}^{\mathfrak{R}}(k, W)$ , first choose independent uniform latent variables  $X_1, \dots, X_k$ . Conditional on them, choose the edge multiplicities  $M_{ij}$  independently with law  $W(X_i, X_j)$ .

Let  $F$  be a finite loopless multigraph on  $[k]$ . Write  $E_+(F)$  for the set of unordered vertex-pairs having positive multiplicity and  $r_e \geq 1$  for the multiplicity of  $e \in E_+(F)$ . Then conditional independence gives the basic identity

$$t(F, W) = \mathbb{E} \prod_{e \in E_+(F)} M_e^{r_e}. \quad (1)$$

Indeed, conditional on  $X_1, \dots, X_k$ , both sides equal the product of the  $r_e$ th moments of the corresponding edge laws. If  $\mathbf{B}_p$  denotes the two-vertex multigraph with  $p$  parallel edges, then

$$t(\mathbf{B}_p, W) = \mathbb{E} M_{12}^p. \quad (2)$$

Thus node-and-edge densities are precisely the mixed polynomial moments of the finite sampled edge vector.

Throughout, every  $W_n$  is assumed to have finite bond moments of every order. Weak convergence of sample laws is ordinary weak convergence of probability measures on the discrete Polish space  $\mathbb{N}_0^{\binom{k}{2}}$ .

### 3 Proof intuition

There are two directions, and each source obstruction has one standard probabilistic cure.

For sampling-to-density convergence, weak convergence already handles every bounded test. A density is an unbounded mixed monomial, so the missing ingredient is uniform integrability. Generalized Hölder reduces the  $L^2$  norm of any such mixed monomial to finitely many one-edge moments. Consequently, once the bond moments converge, every density converges; no regularity or cut-metric argument is needed.

For density-to-sampling convergence, the first bond moment makes each finite family of sample laws tight. All subsequential limits have the same mixed moments, because higher bond moments again give uniform integrability. Marginal Stieltjes determinacy then invokes the half-line version of Petersen's theorem: determinate coordinate marginals make the whole joint law determinate. A uniform exponential moment supplies both uniform integrability and Carleman's determinacy criterion, yielding a clean two-way equivalence. The original question remains broader for arbitrary moment-indeterminate families and for quotient-valued graphons without a chosen probability-kernel representative.

### 4 The moment bridge

**Lemma 1** (Mixed-monomial uniform integrability). *Let  $Z_n = (Z_{n,1}, \dots, Z_{n,q})$  be nonnegative random vectors. Fix positive integers  $r_1, \dots, r_q$ . If*

$$\sup_n \mathbb{E} Z_{n,j}^{2qr_j} < \infty \quad (1 \leq j \leq q),$$

*then  $Y_n = \prod_{j=1}^q Z_{n,j}^{r_j}$  is uniformly integrable.*

*Proof.* Generalized Hölder with all exponents equal to  $q$  gives

$$\mathbb{E} Y_n^2 = \mathbb{E} \prod_{j=1}^q Z_{n,j}^{2r_j} \leq \prod_{j=1}^q \left( \mathbb{E} Z_{n,j}^{2qr_j} \right)^{1/q}. \quad (3)$$

The right side is uniformly bounded. Hence  $(Y_n)$  is bounded in  $L^2$  and is uniformly integrable.  $\square$

**Theorem 2** (Sampling/density moment bridge). *Let  $(W_n)$  be probability multigraphons with all finite bond moments.*

- (i) Suppose  $\mathcal{L}(\mathbb{G}^R(k, W_n))$  converges weakly for every  $k$ . Then  $(W_n)$  is node-and-edge density convergent if and only if

$$\lim_{n \rightarrow \infty} t(\mathbf{B}_p, W_n)$$

exists and is finite for every  $p \geq 1$ .

- (ii) Suppose  $t(F, W_n)$  converges for every finite loopless multigraph  $F$ , and put

$$m_p = \lim_{n \rightarrow \infty} t(\mathbf{B}_p, W_n), \quad m_0 = 1.$$

If  $(m_p)_{p \geq 0}$  is determinate for the Stieltjes moment problem on  $[0, \infty)$ , then  $\mathcal{L}(\mathbb{G}^R(k, W_n))$  converges weakly for every  $k$ .

*Proof.* For (i), necessity is immediate because each  $\mathbf{B}_p$  is itself a test multigraph. Conversely, fix  $F$  on  $[k]$ , enumerate its  $q$  positive edge-pairs as  $e_1, \dots, e_q$ , and let their multiplicities be  $r_1, \dots, r_q$ . The sampled edge vector converges in law by hypothesis. For every  $j$ ,

$$\mathbb{E}(M_{e_j}^{(n)})^{2qr_j} = t(\mathbf{B}_{2qr_j}, W_n),$$

which is uniformly bounded because it converges. Lemma 1 makes  $\prod_j (M_{e_j}^{(n)})^{r_j}$  uniformly integrable. Convergence in law plus uniform integrability, followed by (1), proves convergence of  $t(F, W_n)$ .

For (ii), fix  $k$  and put  $d = \binom{k}{2}$ . Let  $\nu_{n,k}$  be the law of the full  $d$ -coordinate sampled edge vector. Exchangeability and Markov's inequality give, for  $R > 0$ ,

$$\nu_{n,k} \left\{ \max_e M_e > R \right\} \leq \sum_e \frac{\mathbb{E} M_e}{R} = \frac{d t(\mathbf{B}_1, W_n)}{R}. \quad (4)$$

The last numerator is bounded, so  $(\nu_{n,k})_n$  is tight.

Consider any weakly convergent subsequence, with limit  $\nu$ . For a multi-index  $a = (a_e)_{e=1}^d \in \mathbb{N}_0^d$ , apply Lemma 1 to the coordinates with  $a_e > 0$ . The required higher marginal moments are bounded bond densities. Therefore the monomial  $x^a$  is uniformly integrable and

$$\int x^a d\nu(x) = \lim_n \mathbb{E} \prod_e (M_e^{(n)})^{a_e}. \quad (5)$$

The right side is the limiting density of the multigraph on  $[k]$  having multiplicity  $a_e$  on pair  $e$ , so it is independent of the chosen subsequence. In particular, every coordinate marginal of  $\nu$  has moment sequence  $(m_p)$ .

The half-line extension of Petersen's marginal-determinacy theorem [3, 2] says that a measure on  $[0, \infty)^d$  is determined by its mixed moments whenever each coordinate marginal is Stieltjes-determinate. Thus the common mixed moments in (5) determine  $\nu$  uniquely. Every weak cluster point of the tight sequence  $(\nu_{n,k})$  is consequently the same, so the full sequence converges. Since  $k$  was arbitrary, the claim follows.  $\square$

*Remark 3* (A moment-bounded forward criterion). In Theorem 2(i), it is enough to assume

$$\sup_n t(\mathbf{B}_p, W_n) < \infty \quad \text{for every } p \geq 1.$$

Indeed, weak convergence of the two-vertex samples, together with the bounded moment of order  $p+1$ , makes the  $p$ th powers uniformly integrable and hence forces convergence of the  $p$ th bond moments. This recovers the precise uniform-integrability mechanism suggested by the moment-bounded part of the source theory.

## 5 A concrete equivalence regime

**Corollary 4** (Uniform exponential tails). *Suppose there are constants  $\alpha > 0$  and  $C < \infty$  such that*

$$\sup_n \int_{[0,1]^2} \sum_{m=0}^{\infty} e^{\alpha m} W_n(x, y) \{m\} dx dy \leq C. \quad (6)$$

*Then the following are equivalent:*

- (a)  $t(F, W_n)$  converges for every finite loopless multigraph  $F$ ;
- (b)  $\mathcal{L}(\mathbb{G}^R(k, W_n))$  converges weakly for every  $k$ .

*Proof.* Let  $M_n$  have the integrated one-edge law of  $W_n$ . Condition (6) says  $\sup_n \mathbb{E} e^{\alpha M_n} \leq C$ .

If the sample laws converge, then for every  $p$ , the variables  $M_n^p$  are uniformly integrable (for example,  $x^p$  is dominated by a constant multiple of  $e^{\alpha x/2}$ , whose squares have uniformly bounded expectation). Thus their moments converge, and Theorem 2(i) gives density convergence.

Conversely, suppose the densities converge. Since  $x^p \leq p! \alpha^{-p} e^{\alpha x}$  for  $x \geq 0$ , their limiting bond moments satisfy

$$m_p \leq C p! \alpha^{-p}. \quad (7)$$

Consequently

$$\sum_{p=1}^{\infty} m_{2p}^{-1/(2p)} = \infty,$$

because  $((2p)!)^{1/(2p)} \leq 2p$ . Carleman's criterion makes the marginal moment sequence Hamburger-determinate, hence Stieltjes-determinate. Apply Theorem 2(ii).  $\square$

*Remark 5* (Joint exponential control). For a  $k$ -vertex sample with  $d = \binom{k}{2}$  edge coordinates, generalized Hölder also gives

$$\mathbb{E} \exp \left( \frac{\alpha}{d} \sum_e M_e \right) \leq \prod_e (\mathbb{E} e^{\alpha M_e})^{1/d} \leq C.$$

Thus the one-edge hypothesis automatically controls every finite sample jointly; no independence assumption is needed for this estimate.

## 6 Why the hypotheses cannot simply be omitted

**Proposition 6** (Two sharp failure mechanisms). *Neither direction holds without its corresponding moment mechanism.*

- (a) *There are probability multigraphons whose samples converge for every  $k$  but whose node-and-edge densities do not converge.*
- (b) *There are probability multigraphons whose node-and-edge densities all converge but whose two-vertex sample laws do not converge.*

*Proof.* For (a), let  $W_n$  be the constant kernel with edge law

$$\mu_n = \left(1 - \frac{1}{n}\right) \delta_0 + \frac{1}{n} \delta_n.$$

In a fixed  $k$ -vertex sample, the probability that any of the  $d = \binom{k}{2}$  edges is nonzero is at most  $d/n$ . Hence the sample law converges in total variation to the zero multigraph. On the other hand,

$$t(\mathbb{B}_2, W_n) = \int m^2 d\mu_n(m) = n \longrightarrow \infty.$$

For (b), use the distinct probability laws  $\sigma$  and  $\tau$  on  $\mathbb{N}_0$  with all finite moments equal whose existence is recorded explicitly in Examples 4.9 and 4.11 of [1]. Let  $W_{2n}$  be the constant kernel  $\sigma$  and  $W_{2n+1}$  the constant kernel  $\tau$ . For every  $F$ ,

$$t(F, W_n) = \prod_{e \in E_+(F)} \int m^{r_e} d\sigma(m) = \prod_{e \in E_+(F)} \int m^{r_e} d\tau(m),$$

so the density is independent of  $n$ . Yet the two-vertex sample law alternates between the distinct laws  $\sigma$  and  $\tau$ .  $\square$

The second example shows that determinacy is the sharp obstruction to a *general* moments-to-law implication. It does not say that determinacy is necessary for a particular already-convergent sequence.

## 7 Verification, limitations, and novelty status

The proof has four check points. First, (1) is conditional independence applied to a single sampled multiplicity per vertex-pair. Second, (3) uses only marginal moments and therefore does not assume unconditional independence of the sampled edges. Third, (4) gives finite-dimensional tightness before subsequences are taken. Fourth, all mixed moments pass to a subsequential limit by the same  $L^2$  uniform-integrability estimate, after which the cited half-line marginal-determinacy theorem applies. The two counterexamples have also been checked directly.

This packet applies to genuine  $\mathcal{P}(\mathbb{N}_0)$ -valued kernels and to finite multigraphs through their step kernels. It does not solve the measurable selection issue for the source’s quotient-valued moment graphons. Nor does it give a non-tautological necessary-and-sufficient criterion for every individual moment-indeterminate density-convergent sequence. For these reasons the result is classified as a substantial partial answer rather than a full solution of the source’s broad-ended question.

A bounded novelty search on 17 August 2026 used the exact source title and arXiv id, its citation neighborhood, and the phrases “node-and-edge sample convergence”, “multigraphon sampling moment determinate”, “Petersen theorem graphon”, and “Carleman sample convergence graphon”. The closest results found were Kunszenti-Kovács’s Carleman-based uniqueness theorem for Banach-valued graphons [4] and the probability-graphon equivalence of sampling, cut convergence, and homomorphism tests with bounded continuous edge decorations due to Abraham–Delmas–Weibel [5]. The former is a graphon weak-isomorphism theorem; the latter uses bounded  $C_b$  tests, not the unbounded polynomial tests defining node-and-edge multigraph densities. No exact occurrence of Theorem 2 or Corollary 4 was found. Novelty confidence is moderate, not definitive.

**Human-review recommendation.** Check the identification (1), the exponents  $2qr_j$  in (3), and the support hypotheses in the half-line Petersen theorem. If those pass, retain this as a substantial partial theorem giving an exact forward criterion, a determinate reverse theorem, and a concrete two-way equivalence regime.

## References

- [1] D. Kunszenti-Kovács, L. Lovász, and B. Szegedy, *Multigraph limits, unbounded kernels, and Banach space decorated graphs*, J. Funct. Anal. **282** (2022), Article 109284; arXiv:1406.7846, doi:10.1016/j.jfa.2021.109284.
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